

TABLE II
INDIA'S MAJOR
HYDROPOWER
GENERATING UNITS

Name	State	Capacity (MW)
Bhakra	Punjab	1100
Dehar	Himachal Pradesh	990
Nagarjuna	Andhra Pradesh	960
Koyna	Maharashtra	920
Sharavathy	Karnataka	891
Kalinadi	Karnataka	810
Srisailem	Andhra Pradesh	770

electric power production in the country and contribute about 92.5 per cent of the total hydroelectric power produced in the country (refer Table I). Of late, private-sector players have also shown interest in the segment and are planning to build power plants in the Himalayan region and Northeast India.

The government is encouraging the private sector to invest in the hydroelectric segment. Currently, private-sector involvement is only about seven per cent. Private-sector participation will lead to revival of stranded projects. To attract investment, the government is looking at various options like land and other subsidies, and waiving of taxes and custom duties.

Koteshwar hydroelectric power plant

Built on Bhagirathi, a tributary of the river Ganges, Koteshwar Power Project generates about 400MW electricity. It is one of the quickest project implementations in the nation and has been generating almost 1200GWh energy annually.

Led by Tehri Hydro Development Corporation (THDC) India with contract offered to PCL-Intertech Len-Hydro Consortium, the construction

TABLE III
TOP NINE COUNTRIES
(IN TERMS OF CAPACITY)

Country	Power capacity (MWh)	Installed capacity (MW)
Canada	341,312	66,954
USA	319,484	79,511
Brazil	285,603	57,517
China	204,300	65,000
Russia	160,500	44,000
Norway	121,824	27,528
Japan	84,500	27,229
India	82,237	22,083
France	77,500	77,500

work began properly in the year 2007. Prior to this, a river diversion task was completed in 2003. However, the project came to a halt for the next four years following challenges like exponential rise in the cost of raw material leading to fund insufficiency, difficulty in reaching harmonious conclusions with native population of the region and difficult geographic situations. Despite all odds, the first two power generation units of Koteshwar project were commissioned in 2011 and the two more in 2012, completing the project before the expected deadline.

The complete infrastructure of the setup comprises a powerhouse on the right bank of the Bhagirathi river, at the toe of a 97.5-metre high concrete gravity dam. The powerhouse holds the four turbine-power generation units, each having a capacity of 100MW. The generators are semi-umbrella-type vertical shafts connected to a Francis turbine, all manufactured by Bharat Heavy Electricals Limited.

The main reason behind quick completion of the project was changes in managerial procedures and team structures. Time-taking procedures were cut down in the most practical

way to ensure quick procurement and construction. In addition, unconventional methods of constructions were implemented as well. For instance, to erect turbines, crawler cranes were used instead of electrical overhead cranes. Alternate approaches were taken to concrete the generator barrels, excavation, service bay creation and accelerate the reservoir filling.

On commissioning the third and fourth units, the hydroelectric plant started running with full power since March 2012. With a total 400MW generation capacity, the annual energy production by the setup is 1155 gigawatt-hour based on 90 per cent available water.

Operational challenges

Harm to the natural environment and significantly high production cost are the major challenges to the growth of hydroelectric power sector. Creating hydropower plants on an already existing dam is a relatively simpler task. However, to create newer infrastructure, one needs to carefully examine the environmental impact. The Uttarakhand flood disaster is regarded as a result of dam construction at an earthquake-volatile piece of land that aggravated the impact of flood. Construction of a hydropower plant is often considered as a threat to wildlife and water species.

The future potential

To take complete advantage of water, we need to properly define the path for hydropower generation. The bureaucratic process of getting the desired approvals discourages private-sector participation. The concerned authorities must ensure minimal environmental impact of hydropower plants and safe rehabilitation. Foreign participation and attractive schemes from concerned authorities will further help in reaping greater benefits. **EFY**

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